

VASSILY LOMBARD

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Education

Columbia University

BS in Computer Engineering - Richard and Janet Hunter Scholar & Dean's List

Relevant Coursework: Analog Electronics, VLSI Circuits, Solid State Device, Quantum Optics, OS

Sept. 2025 – May 2027

GPA: 3.80 – New York, NY

Earlham College

BA in Data Science - William Roha Mathematics Award Recipient

Relevant Coursework: Artificial Intelligence, Database Management, Data Modeling, Stats & Probability

Sept. 2022 – May 2025

GPA: 3.97 – Richmond, IN

Experience/Leadership

Pixel Design & Characterization Intern

STMicroelectronics

May 2026 - Now

Crolles, France

- Simulated individual sub-blocks of a pixel global shutter test chip using Eldo to identify architectural defaults, bottlenecks and validate system performance.
- Redesigned the SRAM write path architecture within Cadence, successfully minimizing parasitic capacitance at critical nodes to double the maximum operational clock frequency.
- Compiled custom hardware testbenches to characterize physical silicon performance in a cleanroom environment, measuring key metrics including dark current and quantum efficiency.
- Achieved the primary goal of the internship by capturing the sensor's first darkroom image, confirming the innovative aspect of the testchip.

Research Assistant

Columbia University UEIL Lab

Nov. 2025 - May 2026

New York City, NY

- Developed quantitative metrics to analyze ultrasound localization microscopy brain images.
- Engineered a bubble tracking algorithm to reconstruct a map of brain vessels accurate at 86%.

Machine Learning Engineer Intern

Palo IT

May 2025 – Aug. 2025

Paris, France

- Optimized the MoE expert selection architecture in Mixtral, improving load balancing across experts by 9%.
- Implemented a local ChatBot with MCP servers to support different skills reducing documentation writing time by 30%.
- Conceived a RAG based app with AWS Lambda and API Gateway to enhance LLM answers for technical documentation.

Robotics Club President

Earlham College

Sept. 2024 – May 2025

Richmond, IN

- Established and led a student-run robotics club with 20+ members.
- Led coding and robot building workshop using VEX IQ and Arduino platforms.
- Organized and participated in the club's first competition where 5 teams built and programmed robots to follow a line.

Research Assistant

Earlham College

Jan. 2024 – Jul. 2024

Reykjavik, Iceland

- Coded a computer vision model with TensorFlow achieving 74% of accuracy in identifying fish from underwater imagery for fish farm monitoring.

Projects

Happy Hour Finder Web App | Python, AWS Cloud, HTML/CSS

Apr. 2025 – Jul. 2025

- Developed a full-stack FastAPI application to make available happy hour data of bars close to you.
- Collected +100 bars happy hour details using web scraping scripts and Google API.
- Deployed the application with AWS EB Service achieving 50+ monthly visits.

Sign Language Controlled Robot | TensorFlow, OpenCV, Raspberry Pi

Jan. 2025 – May 2025

- Built a Raspberry Pi robot with motor controllers and a camera module for sign language controlled navigation.
- Designed a hybrid fusion computer vision model for static hand sign recognition achieving 70% real time accuracy.
- Created my own dataset of 400+ hand sign images to fine my model for robust classification.

Skills/Interests

Languages: Python, Java, C/C++, Matlab, VBA

Developer Tools: Cadence Virtuoso, Eldo, AWS Cloud, Microsoft Azure (Certified AI-900), Docker, GitHub

Libraires: PyTorch, TensorFlow, OpenCV, FastAPI

Activities and Interests: Competitive golf (1 handicap), Strength training (315 lb bench), Drone Design